

ST/ST/NO/24/101

Mole

A Mole is a unit of Measurement to describe the quantity of atoms in a molecule. This quantity that we are referring, is been discuss in a macroscopic sample containing large amount of atoms or particles. So, invariably, it is important to have a special unit to describe a large number of atoms or particles in a given molecule. But, one important fact to know is that the atoms or particles consist of electrons, ions, protons and other unknown particles which can also be describe as entities. So we can define a Mole (mol): is the amount of a substance that contain as many elementary entities as there are atoms in exactly 12g (or 0.012kg) of the carbon-12 isotope. This number is called Avogadro's number (N_A). The accepted Avogadro's number or value is $N_A = 6.022 \times 10^{23}$.

However, the Avogadro's number is rounded up as $N_A = 6.022 \times 10^{23}$

Supposing a man has a shopping list of items to buy in the Market. The list consist of 2kg of Meat, 2 litres of Gnat oil and one(1) dozen of Eggs. The man remember to add one(1) Mole of salt to the list. The other items can be gotten easily but to get 1 mole of salt will be better imagine, since 1 mole is equal to Avogadro's number $= 6.022 \times 10^{23}$.

Let see how we can get one mole of a salt in the Market?

One(1) mole as a unit of Measurement is not common to ~~the~~ traders, So, let's only chemist or chemistry students that knows the unit.

It is easier to buy one Kg of Salt. or to buy one(1) gramme of salt. In fact, one gramme of salt will not fill a tea spoon. As chemist, we can get 1 mole of salt. First, let ask the basic question, What is a mole mathematically?

A mole is the molecular mass in grammes. So the molecular mass in question is the mass of one molecule of a substance.

⇒ The molecule of our interest is salt with the formula NaCl .

⇒ That, 1 mole of NaCl is the same as 1 mole of molecular mass of NaCl .

⇒ The molecular mass of $\text{NaCl} = 23 + 35 = 58 \text{ grammes}$

⇒ The mass of NaCl in grammes = 58g

~~Salt~~ is not wrong to say, 1 mole of $\text{NaCl}(\text{salt}) = 58 \text{g}$

⇒ When you measure 58g of $\text{NaCl} = 1 \text{ mole of NaCl}$

So let's go further to discuss one mole of a water (H_2O)

⇒ The molecular formula of water = H_2O

⇒ The molecular mass of $\text{H}_2\text{O} = 2 \times 1 + 16 = 2 + 16 = 18 \text{g}$

⇒ That, one mole of water (H_2O) will weigh 18 grammes.

for one mole of a gas molecule, for example oxygen

⇒ The molecular formula of oxygen = O_2

⇒ The molecular mass of oxygen = $2 \times 16 = 32 \text{ grammes}$

So 1 mole of oxygen weighs 32 grammes

Generally, 1 mole of any substances differ from one another due to their molecular mass

We can define a mole in terms of grammes molecular mass (gmm), ~~whereas substances~~

We can also define or describe mole by the number of particles or molecules. These particles are tiny so, if you were given to quantify it, it will be a large quantity of particles or molecules, so since these particles are large, we must estimate it.

⇒ number of molecules = $\frac{\text{Mass of Substance}}{\text{Mass of one molecule}}$

for salt

number of salt molecules = $\frac{\text{Mass of one mole of NaCl(salt)}}{\text{Mass of one molecule}}$

but 1 molecule of hydrogen has a mass of 1.67×10^{-24} .

⇒ That, Mass of NaCl molecule is 58 times heavier or more than Hydrogen

⇒ We can rewrite the formula as

$$\text{Number of NaCl molecules} = \frac{58g}{58 \times 1.67 \times 10^{-24}} = 6.022 \times 10^{23} \text{ molecules}$$

So for one(1) mole of H₂O and Oxygen will give us the same number as Avogadro's Number = 6.022×10^{23} molecules. So we can define moles based on particles or molecules.

When it is a gas, we can define mole in Volume!

⇒ That, what will be the volume occupied by 1 mole of gas?

But, know that 1 mole of a gas occupies 22.4 litres at STP

STP ⇒ Standard Temperature and Pressure

Standard Temperature = 0°C

Standard Pressure = 1 atm

⇒ A room at standard STP will freeze because the temperature is 0°C at 1 atm pressure

Example

What volume of 1 mole of a gas occupy at STP conditions

⇒ It will occupy 22.4 litres of Oxygen at STP and this gives us Molar Volume

So we can make a statement that one mole of Oxygen can be measure by mass = 32g, for molecules or particles = 6.022×10^{23} and by volume = 22.4 L

Summarily, ~~state~~ our definition of a mole shall be: one(1) mole of any substance = 6.022 $\times 10^{23}$ particles

1 mole

can-
be
meas-
ured
by
mass
or
volume

1 Mole
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occupies
22.4 L at STP

Molecular Mass

Once atomic masses of 15 known of a compound, we can calculate the mass of the molecules. The molecular mass sometimes referred as molecular weight is the sum of the atomic mass (in amu) in the molecules.

eg. The molecular mass of H_2O is:

$\Rightarrow 2(\text{atomic mass of H}) + \text{atomic mass of O}$

$\Rightarrow 2(1.008 \text{ amu}) + 16.00 \text{ amu} = 18.02 \text{ amu}$

Question

What is the atomic mass unit of SO_2 ?

Ans

The molecular mass of $SO_2 = 32.07 \text{ amu} + 2(16.00 \text{ amu})$

$$= 32.07 \text{ amu} + 32.00 \text{ amu}$$

$$= 64.07 \text{ amu}$$

Question

What is the molecular mass of caffeine ($C_8H_{10}N_4O_2$)

Ans

$$8(12.01 \text{ amu}) + 10(1.008 \text{ amu}) + 4(14.01 \text{ amu}) + 2(16.00 \text{ amu})$$

$$= 194.20 \text{ amu}$$

From the above understanding that has been established on molecular mass, we can determine the molar mass of a molecule or compound. The molar mass of a compound (in grams) is numerically equal to its molecular mass (amu).

Example, The molecular mass of H_2O is 18.02 amu.

So the molar mass of $H_2O = 18.02 \text{ g}$

Note that 1 mole of H_2O weighs 18.02 g and contains 6.022×10^{23} molecules.

Question

Methane (CH_4) is a natural gas found in the earth. How many moles of CH_4 are present in 4.83 g of CH_4 ?

Ans

The conversion factor needed to convert between grams and moles is the molar mass
 first we must calculate the molar mass of CH_4
 Molar Mass of $\text{CH}_4 = 12.01\text{g} + 4(1.008\text{g})$

$$= 16.04\text{g}$$

$$\Rightarrow 1 \text{ mole of } \text{CH}_4 = 16.04\text{g } \text{CH}_4$$

The conversion factor we need should have grams in the denominator so that the unit grams will cancel, leaving the unit mol in the numerator

$$\Rightarrow \frac{1 \text{ mole } \text{CH}_4}{16.04\text{g } \text{CH}_4}$$

so we now have

$$4.83\text{g } \text{CH}_4 = 1 \text{ mol } \text{CH}_4 \times \frac{16.04\text{g } \text{CH}_4}{16.04\text{g } \text{CH}_4} = 0.301 \text{ mol } \text{CH}_4$$

there we have 0.301 mol of CH_4 in 4.83g of CH_4

Percent Composition of Compounds

The formula of a compound tells us the number of atoms of element in a unit of the compound. We can verify the purity of a compound for use in the laboratory by calculating what percent of the total mass of the compound is contributed by each element from the formula.

The percent composition is the percent by mass of each element in a compound. The percent composition is obtained by dividing the mass of each element in one mole of the compound.

Example

In 1 mole of Hydrogen Peroxide (H_2O_2), there is 2 mole of H atoms and 2 moles of O atoms

The molar mass of H_2O_2 , H and O are: 34.02 (H_2O_2), 1.008 (H) and 16.00 (O) respectively

∴ Percentage composition for each element in H_2O_2 is

$$\%H = \frac{2 \times 1.008}{34.02} \times 100 = 5.92\%$$

$$\%O = \frac{2 \times 16.00}{34.02} \times 100 = 94.08\%$$

The total sum of Percentage = $5.92 + 94.08 = 99.99\% \approx 100\%$

Example

Phosphoric acid (H_3PO_4) is a colorless liquid used in detergent, fertilizers, toothpaste and carbonated drinks. Calculate the Percentage composition by mass of Hydrogen (H), Phosphorus (P) and Oxygen in this compound?

Ans.

The Molar Mass of H_3PO_4 = 97.99

Lat Molar Mass of H = 1, O = 16, P = 30.97

$$\%H = \frac{3(1.008)}{97.99} \times 100 = 3.08\% \quad \text{The total Percentage} = 3.08\% + 31.61\% + 65.31\%$$

$$\%P = \frac{30.97}{97.99} \times 100 = 31.61\% = 100\%$$

$$\%O = \frac{4(16.00)}{97.99} \times 100 = 65.31\%$$

Percent Composition of Compounds

The formula of a compound tells us the number of atoms of element in a unit of the compound for use in the laboratory by calculating what percent of the total mass of the compound is contributed by each element from the formula.

The percent composition is the percent by mass of each element in a compound. The percent composition is obtained by dividing the mass of each element in one mole of the compound.

Example

In one mole of Hydrogen Peroxide (H_2O_2), there is two mole of hydrogen and two mole of oxygen (O) atoms.

The Molar Mass of H_2O_2 , H and O are: 34.02 (H_2O_2), 1.008 (H) and 16.08 (O) respectively

∴ Percentage composition for each element in H_2O_2 is

$$\%H = \frac{2 \times 1.008}{34.02} \times 100\% = 5.94\%$$

$$\%O = \frac{2 \times 16.08}{34.02} \times 100\% = 94.06\%$$

The total sum of percentage = 5.94% + 94.06% = 99.99% $\approx 100\%$

Example

Phosphoric acid (H_3PO_4) is a colourless liquid used in detergents, fertilizers, toothpaste and carbonated drinks. Calculate the percent composition by mass of hydrogen (H), phosphorus (P) and oxygen (O) in H_3PO_4 .

Phosphorus (P) and Oxygen in this compound?

Ans

The molar mass of $\text{H}_3\text{PO}_4 = 97.99$

but molecular mass of H=1, O=16, P=30.97

$$\therefore \% \text{H} = \frac{3(1.008)}{97.99} \times 100 = 3.08\%$$

$$\% \text{P} = \frac{30.97}{97.99} \times 100 = 31.61\%$$

$$\% \text{O} = \frac{4(16.00)}{97.99} \times 100 = 65.31\%$$

The total percentage = $3.08\% + 31.61\% + 65.31\% = 100\%$

Ascorbic acid (Vitamin C) is a compound of 40.92% Carbon (C), 4.58% Hydrogen (H), and 54.50% Oxygen (O) by mass. Determine its empirical formula?

Strategy of Answering this question

In every formula, the subscripts represent the ratio of the number of moles of each element that combine to form one mole of the compound. The question is, how can we have exactly 100g sample of the compound. Do we know the mass of each element in the compound? How do we convert from grams to mole

Ans

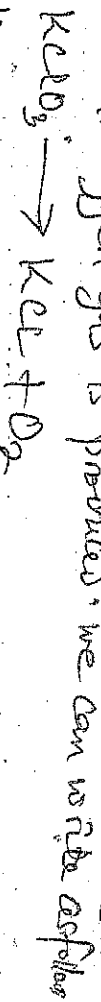
Supposing we have 100g of ascorbic acid, then each percentage can be converted directly to grams, so in this

First, look for elements that appear only once on each side of the equation with the same number of atoms on each side. The formulas containing these elements must have the same coefficient. Therefore there is no need to adjust the coefficients of these elements at this point. Next, look for the elements that appear only once on each side of the equation but in unequal numbers of atoms. Balance these elements, finally, balance elements that appear in two or more formulas on the same side of the equation.

Check your balanced equation to be sure that you have the same total number of each type of atoms on both sides of the equation arrow.

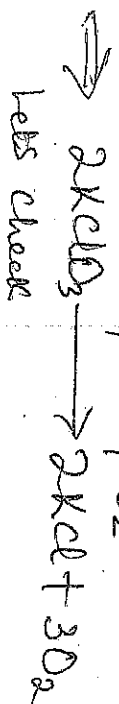
Example

When potassium chlorate is heated in the laboratory a small amount of oxygen gas is produced. We can write the equation



All the three elements (K, Cl, and O) appear only once on each side of the equation but only K and Cl do we have equal numbers of atoms on both sides. That is KClO₃ and KCl must have the same coefficient.

There is three (3) atoms of oxygen on the left and two on the right side. We can balance up by placing 2 in front of KClO₃ and 3 in front of O₂

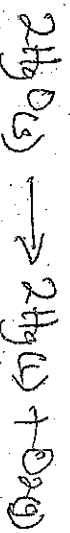


Reactant	Product
K(2)	K(2)
Cl(2)	Cl(2)
O(6)	O(6)

Interestingly, H_2 and O_2 are the reactant while H_2O is the product. Reactants are the starting materials in a chemical equation while products are the substance formed as a result of a chemical reaction.

$\Rightarrow$ Reactant $\longrightarrow$ Products

It is important to provide or indicate physical state of the reactants and product using the letters g, l and s to denote gas, liquid and solid respectively eg



To represent what happens when Sodium chloride ($NaCl$) is added to water (H_2O), we write as



In general, we can balance a chemical equation by the following steps:

① Identify all reactants and product and write their correct formula on the left side and right side of the equation respectively.

② Begin balancing the equation by trying different coefficients to make the number of atoms of each element the same on both sides of the equation. We can change the coefficients (the numbers preceding the formulas) but not the subscripts (that's the numbers within formulas). Changing the subscripts would change the identity of the substance. eg $2NO_2$ means 2 molecules of 'Nitrogen dioxide' but if we double the subscript we will have N_2O_4 which is now a formula of dinitrogen tetra oxide, a completely a different compound.

Chemical Reactions And Chemical Equations

A chemical reaction is a process in which a substance (or substances) is changed into one or more new ~~different~~ substances to express such reactions, we use a standard way to represent chemical equations.

A chemical equation uses a chemical symbol to show what happens during a chemical reaction.

Chemical Equation

When Hydrogen gas (H_2) burns in air which contains Oxygen (O_2), then water (H_2O) is formed. We can use the chemical equation as follows:



Note: The "plus" sign means "react with" and the arrow means "to yield".

The equation in hydrogen reaction with Oxygen is not complete because there are twice as many Oxygen atoms on the left hand side and one Oxygen on the right hand side. We can

balance the equation by placing appropriate coefficient (2 in this case) in front of H_2 and H_2O .

The above equation shows that two hydrogen molecules can combine or react with one molecule of Oxygen (O_2) to form two molecules of water (H_2O). The equation can also read that "2 moles of hydrogen molecules react with 1 mole of Oxygen molecules to produce 2 moles of water molecules".

Further more, we can as well say that 4 g of H_2 react with 32 g of O_2 to give 36 g of H_2O . All these are ways of expressing chemical equations.

of Carbon and Hydrogen in the original 11.5g sample of ethanol as follows.

$$\text{Mass of C} = 32\text{g C}_2 \times \frac{1 \text{ mol C}_2}{44\text{g C}_2} \times \frac{1 \text{ mol C}}{1 \text{ mol C}_2} \times \frac{12.01\text{g C}}{1 \text{ mol C}} = 6.00\text{g C}$$

$$\text{Mass of H} = 13.5\text{g H}_2\text{O} \times \frac{1 \text{ mol H}}{18.02\text{g H}_2\text{O}} \times \frac{2 \text{ mol H}}{1 \text{ mol H}_2\text{O}} \times \frac{1.008\text{g}}{1 \text{ mol H}} = 1.51\text{g H}$$

$\Rightarrow$ 11.5g of Ethanol contains 6.00g of Carbon and 1.51g of Hydrogen. So, the remainder must be Oxygen whose mass is:

$$\begin{aligned} \text{Mass of Oxygen} &= \text{Mass of Sample} - (\text{Mass of C} + \text{Mass of H}) \\ &= 11.5\text{g} - (6.00\text{g} + 1.51\text{g}) \\ &= 11.5\text{g} - 7.51\text{g} \\ &= 4.00\text{g} \end{aligned}$$

$\Rightarrow$ The number of moles of each element present in 11.51g of ethanol is:

$$\text{Moles of C} = 6.00\text{g C} \times \frac{1 \text{ mol C}}{12.01\text{g C}} = 0.50 \text{ mol C}$$

$$\text{Moles of H} = 1.51\text{g H} \times \frac{1 \text{ mol H}}{1.008\text{g}} = 1.5 \text{ mol H}$$

$$\text{moles of O} = 4.00\text{g} \times \frac{1 \text{ mol O}}{16.00\text{g O}} = 0.25 \text{ mol O}$$

So the formula of ethanol will be $\text{C}_{0.50}\text{H}_{1.5}\text{O}_{0.25}$.

Therefore we round up the figure to whole number by dividing each subscript by 0.25, that being the smallest subscript we therefore have an empirical formula of $\text{C}_2\text{H}_6\text{O}$.

3 should be multiply for the formula of Acetic acid

⇒ We shall have $C_3H_4O_3$ as the empirical formula of Acetic acid

Empirical Formulas

In Chemistry, the empirical formula of a chemical compound, is the simplest whole number ratio of atoms present in a compound OR we can say an empirical formula, is the chemical formula of a compound that gives the proportion (ratios) of the element present in a compound but not the actual number or arrangement of atoms, this will be the lowest whole number ratio of the elements in the compound. We can say that empirical formula is a way of expressing chemical formula of a compound in ratio and the percent composition of a compound ~~indirectly~~ leads to its empirical formula.

eg The empirical formula of Acetylene is C_2H , but the molecular formula of Acetylene is C_2H_2 . Glucose has the molecular formula of $C_6H_{12}O_6$ and the empirical formula of Glucose is $C_6H_{12}O_6$

We can always derived a relationship between the molecular formula and the empirical formula - eg Glucose.

Let's consider the compound Ethanol. When ethanol is burned it gives out carbon dioxide (CO_2) and water (H_2O). So, it is obvious that Carbon, Hydrogen and Oxygen is present in ethanol. Suppose that in 11.5g of ethanol was burned to produce 22.0g of CO_2 and 13.5g of H_2O . We can calculate the ^{mass} ~~the~~ _{ratio}

case, there will be 40.92g of C, 4.58g of H and 54.50g of O

The subscripts in the or in a formula represent a mole ratio, we need to convert the grams of each element to ^{moles} moles. The conversion factor needed is the molar mass of each element.

So let's n represent the number of moles of each element so that

$$n_C = 40.92g C \times \frac{1 \text{ mole } C}{12.01g} = 3.407 \text{ mol } C$$

$$n_H = 4.58g H \times \frac{1 \text{ mole } H}{1.008g} = 4.54 \text{ mol } H$$

$$n_O = 54.50g O \times \frac{1 \text{ mole } O}{16.00g} = 3.406 \text{ mol } O$$

from the above, we will arrive at the formula: $C_{3.407}H_{4.54}O_{3.406}$

~~$H_{4.54}O_{3.406}$~~

This above formula gives the identity and mole ratios of atoms present. But, it's important to know that chemical formulas are written in whole numbers. So, let's convert the subscripts in the above formula to whole numbers by dividing all the subscripts by the smaller subscripts which is (3.406)

$$\Rightarrow C = \frac{3.407}{3.406} = 1, H = \frac{4.54}{3.406} = 1.33, O = \frac{3.406}{3.406} = 1$$

This gives us the formula $C_1H_{1.33}O_1$ as the formula for Acetic acid, but we must convert $H_{1.33}$ to a whole number by trial and error procedure.

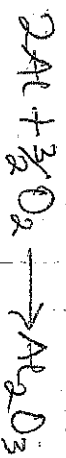
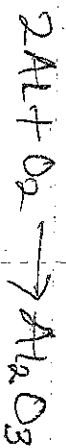
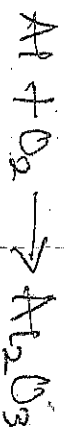
$$\Rightarrow 1.33 \times 1 = 1.33, 1.33 \times 2 = 2.66, 1.33 \times 3 = 3.99 \approx 4$$

So 1.33×3 gives us a whole number by 4. So the subscripts

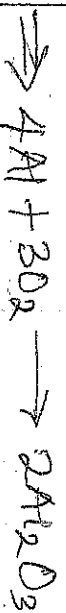
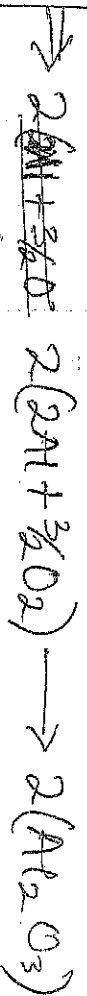
~~Example~~

Example

Balance this equation



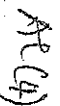
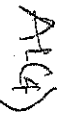
Multiplying both sides by 2



Let's check

Reactants

Products



Percent Composition of Compounds

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Example

In one mole of hydrogen peroxide (H_2O_2), there is two moles of hydrogen and two moles of oxygen (O) atoms.

The molar mass of H_2O_2 , H and O are: 34.02 (H_2O_2), 1.008 (H) and 16.00 (O) respectively.

∴ Percent composition for each element in H_2O_2 is

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